

Chapter 1D: Myocardial Infarction

Paragraph-rich, number-anchored draft for graph-RAG ingestion

Why this matters clinically

Myocardial infarction is the point at which ischemia becomes injury. In stable angina, the myocardium becomes temporarily underperfused, but the cells recover when oxygen supply improves or oxygen demand falls. In myocardial infarction, the ischemic insult is severe enough or prolonged enough to cause myocardial necrosis. This distinction matters because infarcted myocardium does not simply “rest and recover.” It triggers inflammation, electrical instability, loss of contractile function, scar formation, and risk for mechanical rupture.

Clinically, MI is dangerous in two different time frames. In the first minutes to hours, the patient can die from ventricular fibrillation, cardiogenic shock, complete heart block, or ongoing coronary occlusion. Over the next several days, the patient can develop acute heart failure, papillary muscle rupture, ventricular septal rupture, free wall rupture, pericarditis, recurrent infarction, or mural thrombus. Weeks to months later, the patient remains at risk for Dressler syndrome, ventricular aneurysm, chronic heart failure, and recurrent coronary events.

The practical question at the bedside is therefore not only, “Does this patient have an MI?” It is also, “Is there an occluded artery that must be opened now, and what complication am I trying to prevent next?”

Core pathophysiology

Most myocardial infarctions occur when an atherosclerotic plaque ruptures or erodes, exposing thrombogenic material to circulating blood. Platelets adhere to the disrupted plaque surface, become activated, and recruit additional platelets. At the same time, the coagulation cascade generates thrombin and fibrin. The result is a coronary thrombus that can partially or completely obstruct blood flow.

A complete and persistent coronary occlusion tends to produce transmural ischemia, classically creating a STEMI pattern on ECG. A partial occlusion, transient occlusion, or distal embolization may produce subendocardial infarction, often presenting as NSTEMI. This distinction is not absolute in every patient, but it is useful for clinical reasoning: STEMI is treated as an acute coronary occlusion requiring immediate reperfusion, while NSTEMI requires antithrombotic therapy, monitoring, risk stratification, and often early invasive evaluation.

The myocardium is highly oxygen-dependent. When coronary flow drops abruptly, aerobic metabolism fails, ATP levels fall, ion pumps malfunction, and myocytes become electrically unstable. This is why lethal arrhythmias can occur early, sometimes before a large infarct has fully evolved. If ischemia persists, myocyte membranes lose integrity and intracellular proteins such as troponin and CK-MB leak into the blood.

Mechanism Map

Atherosclerotic plaque -> plaque rupture or erosion -> platelet activation and thrombin generation -> coronary thrombus -> partial occlusion causing NSTEMI or complete occlusion causing STEMI.

Numbers to Know: Myocardial Infarction

Topic	Number anchor
Chest pain duration strongly concerning for MI	>20-30 minutes
Troponin begins rising	3-5 hours
Troponin peaks	24-48 hours
Troponin remains elevated	5-14 days
CK-MB begins rising	4-8 hours
CK-MB peaks	About 24 hours

Topic	Number anchor
CK-MB normalizes	48-72 hours
Classic PCI door-to-balloon target	<=90 minutes when PCI-capable
Fibrinolysis benefit highest	Early, especially first 2-3 hours
Fibrinolysis classic window	Up to 12 hours from symptom onset in selected STEMI patients
Mechanical rupture timing	Usually days 3-7
Dressler syndrome timing	Weeks to months after MI

Clinical presentation

The classic MI presentation is prolonged substernal chest pressure that may radiate to the left arm, neck, jaw, back, or epigastrium. Patients often describe the pain as crushing, squeezing, heavy, or like “an elephant sitting on the chest.” The pain commonly lasts more than 20-30 minutes and may not fully resolve with rest or nitroglycerin. Associated symptoms include diaphoresis, dyspnea, nausea, vomiting, weakness, lightheadedness, syncope, and a sense of impending doom.

However, MI does not always look classic. Older adults, diabetic patients, postoperative patients, women, and patients with chronic kidney disease may present with dyspnea, nausea, fatigue, epigastric discomfort, syncope, or acute heart failure rather than obvious chest pain. A patient with diabetes may have autonomic neuropathy that blunts pain perception, creating a silent or painless infarct. This is why MI must remain on the differential when a high-risk patient has unexplained dyspnea, diaphoresis, hypotension, arrhythmia, or pulmonary edema.

For graph-RAG purposes, the important association is: MI leads to myocardial necrosis, myocardial necrosis leads to troponin release, troponin release indicates myocardial injury, myocardial injury increases electrical instability, and electrical instability increases the risk of ventricular tachycardia or ventricular fibrillation.

STEMI versus NSTEMI

STEMI and NSTEMI are both myocardial infarctions, but they differ in the ECG pattern and urgency of reperfusion strategy. STEMI is recognized by ST elevation in contiguous leads or a STEMI-equivalent pattern in the proper clinical context. The initial troponin may still be negative if the patient presents early, so STEMI should not wait for biomarker confirmation. NSTEMI is diagnosed when there is myocardial injury shown by troponin elevation but no persistent STEMI ECG pattern.

NSTEMI often reflects subendocardial injury, whereas STEMI often reflects transmural injury. Subendocardial myocardium is especially vulnerable because it lies farthest from epicardial coronary blood supply and is exposed to the highest intraventricular pressure during systole. This is why global supply-demand mismatch, severe anemia, hypotension, sepsis, or tachyarrhythmia can produce subendocardial ischemia and ST depression. A plaque-rupture NSTEMI is still an ACS emergency, but it is not usually managed with immediate fibrinolysis.

A useful clinical shortcut is: STEMI asks, “Do we open the artery now?” NSTEMI asks, “How high risk is this patient, and how soon do they need invasive evaluation?”

ECG localization

ECG localization connects electrical patterns to coronary anatomy. This is useful because infarct territory predicts complications. Inferior MI may involve the right ventricle or AV node. Anterior MI often involves a larger myocardial territory and can cause pump failure, ventricular aneurysm, or mural thrombus. Posterior MI can be missed if the clinician only looks for ST elevation in the standard leads.

Infarct location	ECG leads	Common vessel	Important association
Inferior	II, III, aVF	RCA most often	Bradycardia, AV block, RV infarct
Anterior/septal	V1-V4	LAD	Large infarct, LV failure, aneurysm
Lateral	I, aVL, V5-V6	LCX or diagonal	May accompany anterior or inferior MI
Posterior	ST depression V1-V3 with tall R waves; posterior leads V7-V9	RCA or LCX	Easy to miss

Infarct location	ECG leads	Common vessel	Important association
Right ventricular	Right-sided leads, especially V4R	Proximal RCA	Hypotension, clear lungs, high JVP

Biomarker reasoning

Troponin is the most important biomarker for myocardial injury. Troponin I and T are released when cardiac myocytes are damaged. In acute MI, troponin usually begins to rise about 3-5 hours after myocardial injury, peaks around 24-48 hours, and may remain elevated for 5-14 days. This makes troponin excellent for diagnosing myocardial injury, but less useful for detecting reinfarction shortly after a recent MI because it stays elevated for so long.

CK-MB is less commonly used now but remains exam-relevant because it returns to baseline faster. CK-MB usually begins to rise about 4-8 hours after infarction, peaks around 24 hours, and normalizes within 48-72 hours. Therefore, if a patient has recurrent ischemic chest pain two days after an MI, CK-MB can help identify reinfarction because a new CK-MB rise after prior normalization is suspicious.

A key graph-RAG relationship is: troponin is a myocardial injury marker, while CK-MB is a reinfarction marker after recent MI because it clears faster.

Marker	Rises	Peaks	Returns/Duration	Best use
Troponin I/T	3-5 hr	24-48 hr	Elevated 5-14 days	Diagnosis of myocardial injury
CK-MB	4-8 hr	~24 hr	Normal 48-72 hr	Suspected reinfarction after recent MI

Immediate management of suspected MI

The initial management of MI begins before every detail of the diagnosis is known. The clinician places the patient on a cardiac monitor, obtains IV access, checks vital signs and oxygen saturation, obtains a 12-lead ECG, and sends troponin and basic labs. If the ECG shows STEMI or a STEMI equivalent, the reperfusion pathway begins immediately.

Aspirin is given early unless there is a true contraindication such as active major bleeding or severe allergy. Classic teaching often uses a chewable loading dose in the 162-325 mg range, and the uploaded cardiology source specifically lists 325 mg in its ACS/MI treatment material. P2Y12 inhibitors such as clopidogrel, ticagrelor, or prasugrel are added depending on the clinical pathway and planned intervention. Anticoagulation with unfractionated heparin, enoxaparin, or another agent is used to prevent thrombus propagation.

Nitrates can relieve ischemic chest pain by reducing preload and myocardial oxygen demand, but they are not harmless. They should be avoided in hypotension, suspected right ventricular infarction, severe aortic stenosis, or recent PDE-5 inhibitor use. Beta-blockers reduce heart rate, blood pressure, and contractility, thereby lowering oxygen demand, but they should not be given early to a patient in cardiogenic shock, acute decompensated heart failure, severe bradycardia, or high-grade AV block. Oxygen is appropriate if the patient is hypoxemic, but routine oxygen in a normally saturated patient does not improve outcomes. High-intensity statin therapy is started early to stabilize plaque and reduce future events.

Reperfusion in STEMI

STEMI management centers on reperfusion. The infarct artery must be opened quickly because myocardium is lost over time. Primary PCI is preferred when it can be performed rapidly by an experienced team. A classic benchmark is door-to-balloon time within 90 minutes at PCI-capable centers. The exact logistics vary by system, but the principle is constant: shorter ischemic time means more salvageable myocardium.

Fibrinolysis is considered when a patient has STEMI within an appropriate symptom window and PCI cannot be performed quickly. Benefit is greatest early, especially within the first 2-3 hours, and is generally considered within 12 hours of symptom onset in selected patients without contraindications. Fibrinolytics are not used for unstable angina or NSTEMI because the risk-benefit balance is different and there is no persistent ST-elevation occlusion pattern. Before giving fibrinolysis, the clinician must screen for contraindications such as prior intracranial hemorrhage, suspected aortic dissection, active bleeding, recent major surgery, significant head trauma, or severe uncontrolled hypertension.

CABG is not the routine immediate treatment for most STEMIs, but it becomes important when coronary anatomy is unsuitable for PCI, when there is left main or complex multivessel disease, when PCI fails, or when mechanical

complications require surgery.

STEMI Reperfusion Pathway

STEMI or STEMI-equivalent pattern -> ask whether PCI can be performed rapidly -> if yes, primary PCI -> if no, check fibrinolysis eligibility -> if no contraindications and within window, give fibrinolysis and transfer for angiography or rescue PCI if needed.

Right ventricular infarction

Right ventricular MI is an important special case because the treatment instincts for left-sided pulmonary edema can harm the patient. RV infarction usually occurs with inferior MI due to proximal RCA occlusion. The right ventricle becomes unable to pump blood effectively into the pulmonary circulation, so left ventricular preload falls. These patients can become hypotensive despite clear lungs.

The classic clue is inferior MI plus hypotension, elevated JVP, and clear lungs. Right-sided ECG leads, especially V4R, can help confirm the diagnosis. Because RV infarct patients are preload-dependent, nitrates and aggressive diuresis can cause cardiovascular collapse. Management emphasizes maintaining preload, avoiding unnecessary preload reduction, and restoring coronary perfusion.

Graph-RAG association: inferior MI points toward proximal RCA involvement; proximal RCA involvement can cause RV infarct; RV infarct causes hypotension, JVD, and clear lungs; this pattern means avoid nitrates and diuretics and consider cautious fluids if needed.

Post-MI medications

After the acute event, medications are used to prevent reinfarction, reduce remodeling, lower mortality, and control symptoms. Aspirin remains foundational unless contraindicated. A P2Y12 inhibitor is added for dual antiplatelet therapy, especially after stenting or ACS. High-intensity statin therapy is continued because plaque stabilization and LDL lowering reduce recurrent events.

Beta-blockers are particularly useful after MI when there is reduced EF, arrhythmia risk, or ongoing angina, although patient selection matters. ACE inhibitors or ARBs are especially important in patients with anterior MI, reduced EF, hypertension, diabetes, or chronic kidney disease because they reduce maladaptive remodeling. Mineralocorticoid receptor antagonists may be used in selected patients with reduced EF and either diabetes or heart failure symptoms if renal function and potassium allow.

Medication group	Why it matters after MI	Major cautions
Aspirin	Prevents platelet-mediated coronary thrombosis	Major bleeding, allergy
P2Y12 inhibitor	Dual antiplatelet therapy after ACS/stent	Bleeding, surgical timing
High-intensity statin	Plaque stabilization and event reduction	Liver injury, myopathy
Beta-blocker	Lowers demand and arrhythmia risk	Shock, acute HF, bradycardia, AV block
ACE inhibitor/ARB	Reduces remodeling, especially anterior MI or reduced EF	AKI, hyperkalemia, pregnancy

Post-MI complications by timing

The timing of complications is one of the most testable parts of MI. In the first minutes to hours, arrhythmias dominate. Ventricular tachycardia and ventricular fibrillation are major causes of early death. Bradyarrhythmias and AV block are especially associated with inferior MI because the AV node is often supplied by the RCA. Sinus bradycardia may occur early in inferior MI and can be observed if mild, but severe symptomatic bradycardia may require atropine or pacing.

Over the next 1-3 days, acute pericarditis can occur due to inflammation overlying the infarcted myocardium. Patients develop pleuritic chest pain that may improve when leaning forward. This early post-MI pericarditis is distinct from Dressler syndrome, which occurs weeks to months later and is immune-mediated.

Around days 3-7, macrophages digest necrotic myocardium and the infarcted wall is structurally weak. This is the danger period for mechanical rupture. Free wall rupture causes hemopericardium and cardiac tamponade, often with sudden pulseless electrical activity. Ventricular septal rupture causes a new harsh systolic murmur with acute heart failure or shock. Papillary muscle rupture causes acute severe mitral regurgitation, pulmonary edema, and a new systolic murmur. These complications are surgical emergencies.

Weeks to months later, the patient may develop Dressler syndrome, ventricular aneurysm, mural thrombus, or chronic heart failure. Dressler syndrome is an autoimmune post-cardiac injury syndrome with fever, malaise, pleuritic chest pain, and pericarditis. Ventricular aneurysm can cause persistent ST elevation, heart failure, ventricular arrhythmias, and mural thrombus.

Timing after MI	Complication	Classic clue
Minutes to hours	VT/VF	Sudden cardiac death
First 24 hours	Arrhythmias, acute pump failure	Palpitations, syncope, pulmonary edema
1-3 days	Fibrinous pericarditis	Pleuritic pain, friction rub
3-7 days	Papillary muscle rupture	Acute MR, pulmonary edema
3-7 days	Ventricular septal rupture	New harsh systolic murmur, shock
3-7 days	Free wall rupture	Tamponade, PEA arrest
Days to weeks	Mural thrombus	Embolic stroke risk, especially anterior MI
Weeks to months	Dressler syndrome	Fever, pericarditis, pleuritis
Weeks to months	Ventricular aneurysm	Persistent ST elevation, HF, arrhythmias

High-yield illness associations

Finding	Association
Inferior STEMI + bradycardia	RCA involvement affecting AV node
Inferior STEMI + hypotension + clear lungs	Right ventricular MI
Anterior STEMI	LAD occlusion, larger LV territory
Recurrent pain after recent MI	Reinfarction; check CK-MB trend
New systolic murmur + pulmonary edema	Papillary muscle rupture causing acute MR
New harsh systolic murmur + shock	Ventricular septal rupture
PEA arrest several days after MI	Free wall rupture with tamponade
Fever + pleuritic pain weeks after MI	Dressler syndrome
Persistent ST elevation weeks after MI	Ventricular aneurysm
Stroke after large anterior MI	LV mural thrombus embolization

Common pitfalls

One common mistake is waiting for troponin before treating STEMI. Troponin may still be negative early, but the ECG can already show an occlusive infarction pattern. Another mistake is treating all chest pain after MI as recurrent ACS without considering mechanical complications. A new murmur, shock, pulmonary edema, or sudden tamponade physiology after MI should immediately raise concern for rupture syndromes.

A third pitfall is giving nitrates or diuretics to an RV infarct patient with hypotension. In left-sided pulmonary edema, preload reduction may help. In RV infarction, preload reduction can collapse cardiac output. A fourth pitfall is using troponin to diagnose reinfarction shortly after a known MI. Because troponin remains elevated for days, CK-MB is more useful if it had returned toward baseline and then rises again.

Finally, clinicians must avoid reflex anticoagulation or fibrinolysis if the presentation could be aortic dissection. Chest pain radiating to the back with pulse deficits, neurologic symptoms, or mediastinal widening should trigger dissection evaluation before therapies that increase bleeding risk.

Practice questions

Question 1

A 63-year-old man presents with 50 minutes of crushing substernal chest pain and diaphoresis. ECG shows ST elevation in leads V2-V5. His first troponin is normal.

Answer: Anterior STEMI; activate reperfusion pathway immediately. Troponin can be normal early because it may take 3-5 hours to rise. ST elevation in V2-V5 suggests LAD territory infarction, and STEMI is treated based on ECG and clinical presentation rather than waiting for biomarkers.

Question 2

A 71-year-old woman develops recurrent chest pain two days after an NSTEMI. Her troponin remains elevated from the original event.

Answer: Check CK-MB trend for reinfarction. Troponin can remain elevated for 5-14 days, making it hard to interpret soon after MI. CK-MB returns to baseline within 48-72 hours, so a new rise can support reinfarction.

Question 3

A patient with inferior STEMI becomes hypotensive. Lung exam is clear, and JVP is elevated.

Answer: Suspect right ventricular infarction; avoid nitrates and aggressive diuretics. Inferior MI can involve the proximal RCA and right ventricle. RV infarct patients are preload-dependent, so preload-reducing drugs can worsen hypotension.

Question 4

Five days after MI, a patient develops sudden pulmonary edema and a new loud systolic murmur.

Answer: Papillary muscle rupture causing acute mitral regurgitation. Mechanical rupture complications classically occur around days 3-7, when macrophage digestion weakens necrotic tissue. Papillary muscle rupture causes acute MR and pulmonary edema.

Question 5

Several weeks after MI, a patient develops fever, malaise, pleuritic chest pain, and a friction rub.

Answer: Dressler syndrome. Dressler syndrome is an immune-mediated post-MI pericarditis that occurs weeks to months after infarction.

Chapter summary

Myocardial infarction is myocardial necrosis due to prolonged ischemia. STEMI is an ECG-defined reperfusion emergency, while NSTEMI is diagnosed by ischemic presentation with troponin elevation but no persistent STEMI pattern. Troponin begins rising in about 3-5 hours, peaks at 24-48 hours, and remains elevated for 5-14 days. CK-MB rises in 4-8 hours, peaks around 24 hours, and normalizes in 48-72 hours, making it useful for suspected reinfarction.

ECG localization links infarct territory to coronary anatomy and complications: inferior MI suggests RCA and possible AV block or RV infarct, anterior MI suggests LAD and larger LV injury, and posterior MI can be missed without posterior leads. STEMI requires rapid reperfusion, preferably PCI when available quickly, while NSTEMI requires antithrombotic therapy, monitoring, and risk-based invasive evaluation.

Post-MI complications follow a timeline: arrhythmias early, pericarditis after 1-3 days, rupture syndromes around days 3-7, and Dressler syndrome weeks to months later. These time-linked associations are especially important for clinical reasoning, exam recognition, and graph-RAG concept mapping.

Source and originality note

This chapter is original Medvale explanatory writing. The uploaded cardiology excerpt was used as a coverage checklist for MI symptoms, ECG localization, biomarker timing, acute treatment, reperfusion, right ventricular infarction, and post-MI

complications. The text is not a rewrite of the source; it is structured for clinical reasoning and downstream graph-RAG extraction.